

INVESTIGATION AND EVALUATION OF SECONDARY MACHINING PROCESSES.

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Abstract.

This report investigates six key secondary machining processes; three traditional (centre lathe turning, milling, pillar drilling) and three specialist (broaching, honing, wire EDM), by analysing six corresponding engineering components. For each process, it explains how it functions, why it was selected, the manufacturing sequence used to produce specific features, the accuracy achieved, batch size suitability, and sustainability considerations (energy use, cutting fluids, waste management). Comparative tables summarise performance and justify process selection. The report concludes that traditional machining is versatile for general features in low to medium batches, while specialist methods are essential for complex geometries and high-precision requirements.

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Introduction.

Secondary machining processes are used to refine and finish parts after initial shaping to achieve final dimensions, features, and surface quality required by engineering designs. These processes ensure functionality, fit, and performance in manufactured components. Traditional methods (turning, milling, drilling) apply mechanical cutting, whereas specialist methods (broaching, honing, wire EDM) use advanced cutting or material removal strategies to achieve greater complexity or tighter tolerances. The purpose of this report is to examine these processes through real component case studies, explain how they are used, and evaluate why each method was chosen based on accuracy, batch size, and sustainability.

Background Theory.

Machining involves removing material to shape workpieces into engineered parts. Material removal occurs through mechanical cutting (traditional) or other mechanisms such as abrasion (honing) or electrical discharge (EDM).

1. Accuracy and tolerance: defines acceptable errors from nominal dimensions, tighter tolerances often demand advanced specialist processes.
2. Batch size: affects process choice, manual or traditional approaches suit smaller batches, specialist tools justify cost in medium or large runs.
3. Sustainability: Includes energy consumption, cutting fluid use, and recyclable waste.
4. Surface roughness(Ra) : is the microscopic peaks or valleys on a material surface caused by manufacturing processes like grinding.

Case studies - Traditional Secondary Processes.

Centre lathe turning - Aluminium part.

Component Description:

Aluminium part with tapered ends, drilled and tapped holes, and a knurled surface.

Material and selection:

Aluminium alloy: chosen because it is lightweight, corrosion resistant and easy to machine with minimal tool wear.

Machine overview:

A centre lathe rotates the workpiece while a stationary single-point tool removes material. Originating from early metalworking, it is fundamental for cylindrical features.

Manufacturing sequence.

1. Mount bar stock in three-jaw chuck.
2. Face end surfaces to datum.
3. Perform rough turning to approximate diameters.
4. Machine taper using compound slide.
5. Apply a knurling tool for texture.
6. Centre drill and drill holes.
7. Tap threads in holes.
8. Inspect with micrometers and gauges.

Accuracy and tolerance: ± 0.05 – ± 0.10 mm achievable.

Batch suitability: low to medium batch.

Reasons:

Rapid Setup: Manual lathes allow for fast tool changes and manual adjustments, making them more cost-effective than programming a computer for only a few parts. Operators can make instant tweaks to the cut without stopping to rewrite code, which is ideal for experimental or one-off pieces.

Sustainability.

Recyclable swarf; Aluminium swarf retains their material properties and can be melted down repeatedly without losing quality.

- Energy efficiency; Recycling these chips uses 95% less energy than making new aluminium from raw ore, significantly lowering the carbon footprint of the project.

Low coolant use; Thermal Conductivity: Aluminium naturally pulls heat away from the tool and into the chip. This allows the machine to run with minimal lubrication.

- Reduced Waste: Using less coolant means there is less hazardous fluid to dispose of and the metal chips stay cleaner, making them easier and cheaper to recycle.

Other cutting parameters include, spindle speed(rev.per.minute), feed rate and depth of cut. These are all necessary to prevent gumming(metal melting unto tool), ensure smooth finish, control heat generating and chip formation, accuracy and piece quality.

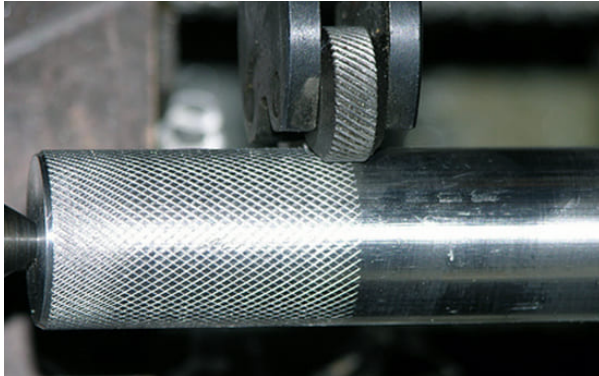


*a pic of a work in progress on a centre

lathe.*pic from google.

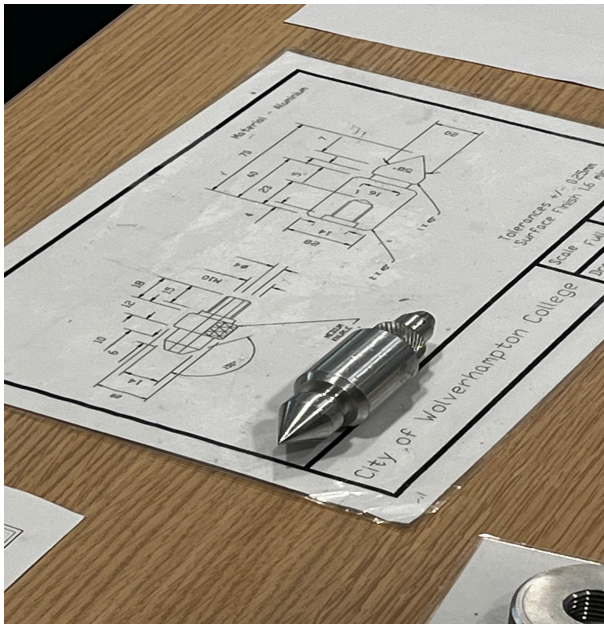


* similar piece of aluminium done on the lathe.



* piece being

knurled on the lathe using a knurling tool.*2nd pic from google.



*a pic of the finished piece.



* the

work piece being machined on the lathe.



* me machining the piece on the lathe.

Milling - Mild Steel block.

Component description.

Long mild steel block with multiple holes of differing locations, a see-through curved pocket, and a deep cut at the edge.

Material and selection.

Mild steel: strong, versatile and common in structural components.

Machine overview.

Milling machines use multi-tooth rotating cutters on controlled axes to remove material. Historically developed during industrialisation for flat and complex surfaces. It is often preferable for one-off prototypes since it avoids the time-consuming setup and programming required for CNC milling, making it faster and more cost effective for small scale tasks.

Manufacturing sequence.

1. Secure works on the milling table using vice and parallels.
2. Face mill top surface.
3. Spot drill and drill holes.
4. Boring or reaming to achieve final hole dimensions.
5. Contour milling for curved pockets.
6. Slot milling for deep edge cuts.
7. Deburr and inspect.

Accuracy and Tolerance: ± 0.05 mm with good fixturing.

Batch suitability: small to medium.

Reasons:

allows for flexible setup changes without the high overhead of mass-production tooling, making it cost-effective for custom or limited runs.

Sustainability.

Recyclable steel swarf is collected for scrap value. Steel swarf retains its fundamental material properties and is widely recycled in electric arc furnaces. This reduces the demand for virgin iron ore and the intensive processing it requires. Also in terms of energy efficiency, recycling steel chips uses

approximately 60-75% less energy than producing new steel from raw ore. This substantial energy saving directly lowers the project's carbon footprint. Also coolant reuse is recommended. Mild steel requires more cooling than aluminum. Using targeted delivery systems reduces fluid consumption, minimizing hazardous waste and producing cleaner, more recyclable chips.

Other critical cutting parameters that ensures quality alongside sustainability includes;

Spindle Speed and Feed Rate: These are balanced to manage heat generation, prevent work hardening of the steel surface, and promote the formation of manageable chips.

Depth of Cut: This is optimized to maintain accuracy, achieve a smooth surface finish, and control cutting forces. Correct parameters extend tool life, reduce tool waste, and ensure efficient material removal with minimal energy waste. Proper management of these factors results in a high-quality part, longer tool life, cleaner and more recyclable waste streams, and lower overall energy consumption.



* milling cutter and tools (pic from google).



*the steel block being machined on the mill.



* work in

progress/ cutting tools.



*machined work

piece.

Pillar Drilling - Aluminium Holework.

Component description.

Rectangular aluminium part with four holes of varied sizes, counterbore, countersink, tapped threads, and polished surface.

Material and selection.

Aluminium for easy drilling and polishing.

Machine overview.

Pillar drill/ drill press clamps and feeds rotating drills vertically. A simple but accurate machine for holemaking.

Manufacturing sequence.

1. Mark holes centres and centre drill.
2. Drill holes progressively.
3. Perform counterboring and countersinking.
4. Tap thread with the appropriate tool and size.
5. Polish surface.
6. Inspect hole locations and finish.

Accuracy and Tolerance: around ± 0.05 mm for hole positions.

Batch suitability: one-off and small batches.

The reason is, the pillar drill is ideal for limited runs and prototypes due to its straightforward setup and minimal tooling requirements, avoiding the high costs of automated systems.

Sustainability.

Recyclable Swarf: Aluminium swarf is highly valuable and should be collected cleanly. It can be recycled repeatedly without loss of quality, significantly reducing the need for energy-intensive primary aluminium production.

Energy Efficiency: Recycling aluminium swarf uses approximately 95% less energy than producing new metal from ore, drastically cutting the carbon footprint of the project.

Reduced Coolant Use: Aluminium's excellent thermal conductivity allows for operations with minimal lubrication. Using a mist system or air blast instead of flood coolant reduces fluid consumption and disposal, keeping chips clean and enhancing their recycling value.

For a pillar drill machining aluminium, key parameters must be controlled to ensure efficiency and part quality. Some of these parameters include;

Drill Bit Speed (RPM): A higher speed is typically used but must be controlled to prevent the aluminium from melting and gumming onto the drill bits.

Feed Rate: A consistent and appropriate feed pressure is crucial to allow the drill to cut cleanly, promote effective chip ejection, and avoid excessive heat buildup. Also, using sharp bits with polished flutes designed for aluminium prevents material adhesion, ensures smooth hole finishes, and extends tool life.



* a

countersink tool in hand and a tapping tool being used to tap a hole.



* drilling in

progress and a counterbore tool in hand.



*tools being used and the workpiece in progress.



and a pic of a pillar drill.



*counterbore in action

Case Studies - Specialist Secondary Processes.

Broaching - Gear (toothed wheel).

Component description.

A toothed wheel which has uniform tooth profiles. This is done via external broaching.

Material and selection.

Alloy or mild steel for strength and wear resistance.

Machine Overview.

Broaching machines push/pull a multi-toothed broach tool through the work to produce accurate profiles. Broaching originated in the early 20th century for efficient profile cutting.

Manufacturing sequence.

1. Prepare a pre bore.
2. Mount blank in broaching machine.
3. Broach passes through cutting sequentially.
4. Deburr and inspect tooth profiles.

Accuracy and Tolerances: ± 0.01 – ± 0.05 mm.

Batch suitability: medium to large production.

Reasons; Broaching is ideal for medium to high production runs due to the high initial cost and complexity of the broach tool. This investment is justified by exceptional consistency, high throughput, and superior surface finish for gear teeth, making it cost-effective for significant quantities.

Sustainability.

Recyclable Swarf: Steel or alloy swarf generated during broaching is collected for recycling. The material retains its properties and is reprocessed in furnaces, reducing the demand for virgin ore and the associated mining and refining impacts.

In terms of energy efficiency, recycling this steel scrap uses approximately 60-75% less energy than primary production, substantially lowering the project's carbon footprint over many parts.

Also, effective fluid management is critical in broaching to manage heat, ensure finish quality, and extend the life of the expensive broach tool. Using filtered and recirculated coolant systems minimizes waste. Alternatively, Minimum Quantity Lubrication (MQL) can be highly effective, reducing fluid consumption and yielding cleaner, drier chips for recycling.

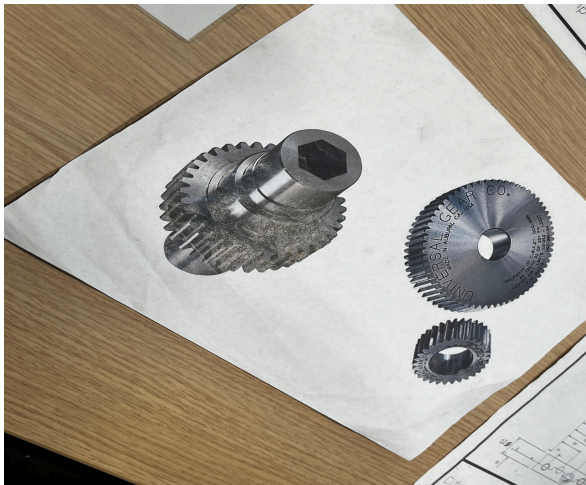
Some key broaching parameters that are essential to this task includes;

Broaching Speed: Optimized to balance production rate with controlled heat generation and to prevent premature tool wear or surface damage on the gear teeth.

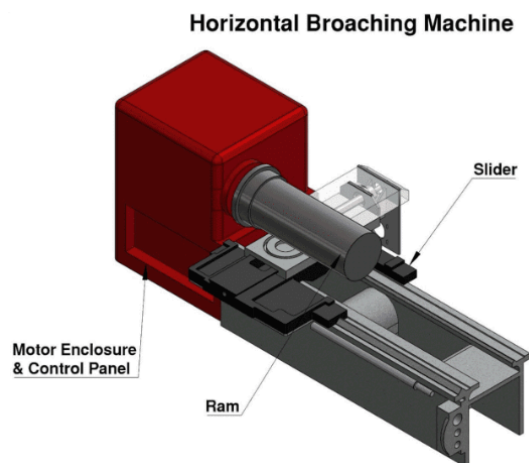
Stroke Length and Setup: Precisely configured for the gear profile to ensure accuracy and consistency across the batch, minimizing material and energy waste from rejected parts.

Broach Tooth Design and Condition: The geometry, pitch, and sharpness of the broach teeth are necessary. A well-maintained tool produces better surface finishes, generates predictable chips, and operates with lower energy consumption per part.

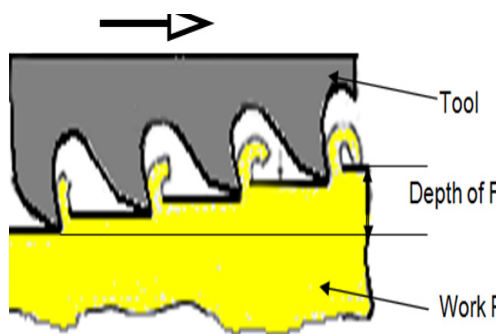
Proper management of these factors results in high-precision gears, maximized broach tool life, reduced waste, and efficient material use, aligning high-volume manufacturing with sustainable practice.



* a picture of the task at hand.



* a well labelled diagram of the broaching machine.(from google).



* manner in which it functions.(from google)



*gears(toothed wheel) (pic from google).

Honing - Cylinder head or Engine bore components.

Component Description.

Cylinder head bores need precise surface finish and geometry.

Material and selection.

Aluminium alloy or cast iron - typical automotive head materials.

Machine overview.

Honing machines use reciprocating abrasive stones to refine internal bores beyond what drilling or boring can achieve.

Manufacturing sequence.

1. Pre- bore cylinder if necessary.
2. Select correct honing stones and machine settings.
3. Perform honing passes to achieve final diameter and surface pattern.
4. Inspect with bore gauges and surface finish tools.

Accuracy and Tolerance: Tight (often $<\pm 0.01$ mm).

Batch suitability: medium to high volume.

Reasons; Honing is a critical finishing process in engine manufacturing. While it can be adapted for prototypes, the setup for consistent bore geometry, surface finish, and straightness is most efficient and cost-effective in production runs, aligning with the high-volume nature of cylinder head production.

Sustainability.

Recyclable Waste Streams: The fine ferrous (or aluminium) particles removed during honing are captured in the coolant filtration system. This metal sludge can be processed to recover material for recycling, diverting it from landfill.

In terms of energy efficiency as a final, precise finishing operation, honing ensures parts meet specification without rejection. This process efficiency, combined with the high recyclability of the base metal, supports the overall energy savings of using recycled engine components.

NB: Honing produces an abrasive sludge rather than the known chips because the process uses extremely fine grit sizes that only removes microscopic amounts of material. This mixing with large volumes of honing fluids form the sludge.

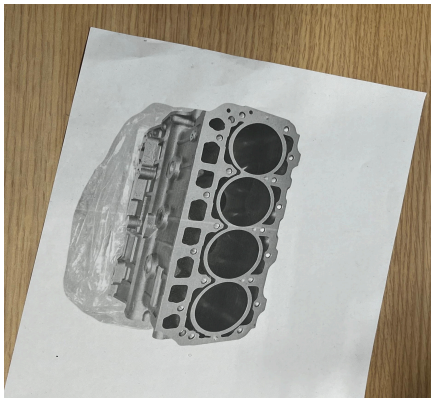
Coolant Filtration and Reuse: Honing relies on a clean, pressurized coolant (honing oil) to flush away particles, control temperature, and extend stone life. Using a closed-loop system with integrated filtration and chip removal is standard. This maximizes coolant reuse, minimizes fluid consumption and disposal, and keeps the waste metal clean for recovery.

Other key parameters essential to ensure the best quality produce includes:

Honing Stone Pressure and Speed: These are balanced to achieve the required material removal rate and surface finish (plateau profile) without introducing excessive heat or causing glazing of the abrasive stones.

Abrasive Selection and Conditioning: The choice of stone grit, bond, and proper dressing regimen is critical. Optimized abrasives deliver the specified finish efficiently, maximize stone life to reduce consumable waste, and generate manageable swarf for the filtration system.

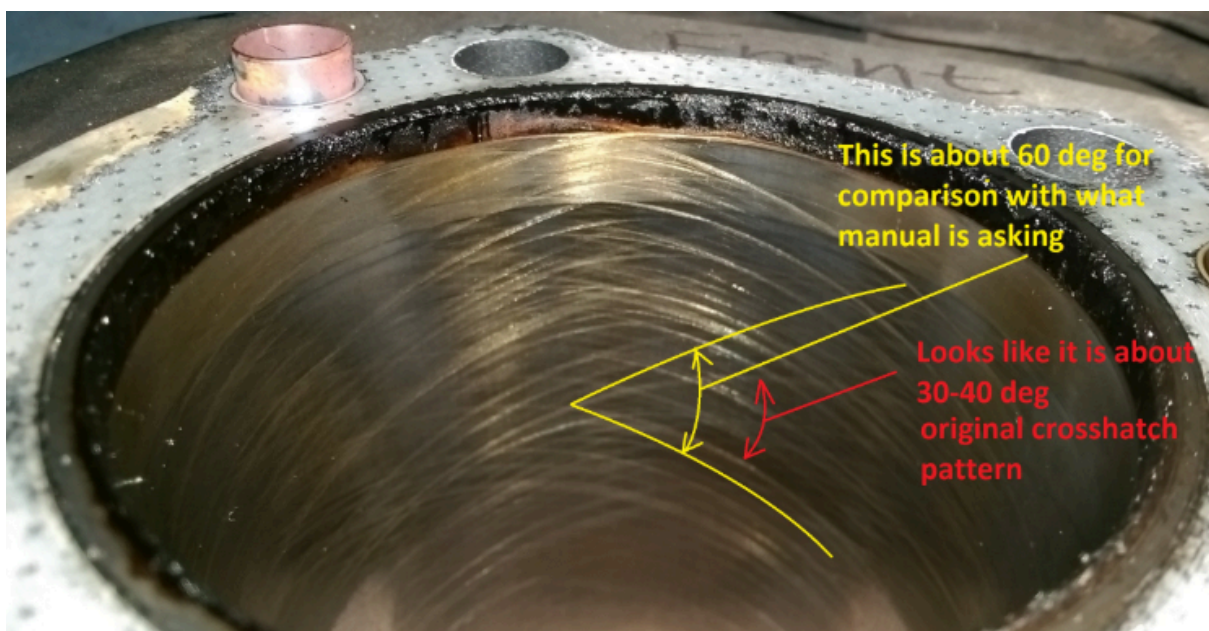
Proper management of these factors results in producing the best quality.



* a pic of the car cylinder head.



* honing stones in bore.(pic from google)



*cross hatch surface pattern. (pic from google).

Wire EDM - Snowflake block.

Component Description.

Intricate snowflake- shaped metal block requiring high precision.

Material and Selection.

Hardened tool steel or similar - chosen for wear resistance and stability.

Machine overview.

Wire EDM uses a thin, continually fed wire and dielectric fluid to erode material via electrical discharges. Ideal for intricate contours.

Manufacturing Sequence.

1. Program CAD path of the snowflake outline.
2. Mount workpiece in EDM machine.
3. Flush dielectric fluid and begin wire cutting.
4. Complete cut and inspect profile accuracy.

Accuracy and Tolerance: Very high; typically ± 0.02 mm.

Batch suitability: prototype to medium volume.

Reasons; Wire EDM is uniquely suited for intricate, delicate shapes like a snowflake, regardless of material hardness. It is ideal for one-off prototypes, tools, and low to medium batches where the complexity of the design outweighs its slower cutting speed. The high initial setup and programming time is justified by unparalleled precision and the ability to produce fragile geometries without mechanical stress.

Sustainability.

Recyclable Material: The main metal block (the slug) remains unpolluted and fully recyclable post-cut. The fine metallic particles (eroded from the cut path) suspended in the dielectric fluid can be captured by filtration systems and processed for material recovery.

In terms of energy efficiency, while the process itself is energy-intensive per hour of operation, its ability to produce finished, complex parts in a single operation, eliminating multiple roughing and finishing steps, can lead to net energy savings. It also prevents the waste of material that would be removed to create such intricate internal features by conventional means.

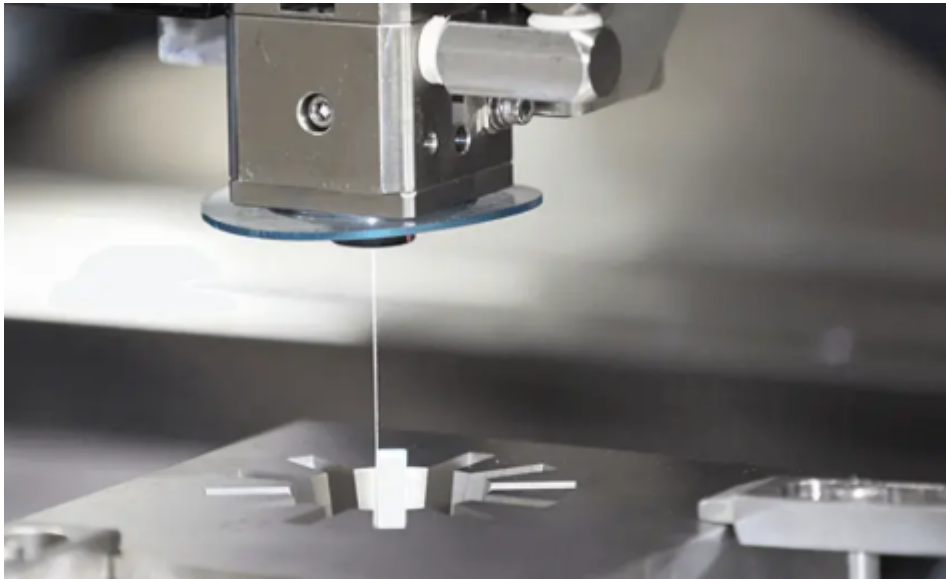
Dielectric Fluid Management: Using deionised water as the dielectric is standard. Effective closed-loop systems with filtration, chilling, and resin purification are critical. This extends fluid life indefinitely, minimizes water consumption, and continuously removes contaminating particles, ensuring cutting stability and reducing waste disposal.

Other critical parameters for quality includes:

Wire Tension and Speed: Higher tension and controlled feed are vital for maintaining straightness and precision in intricate contours, preventing deviations that could break delicate snowflake points.

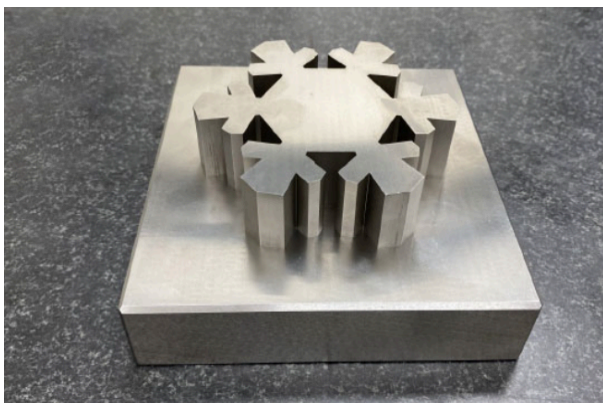
Flushing and Filtration: Excellent dielectric flow through the narrow kerf is essential to remove eroded particles and prevent secondary discharges

(arcing). Efficient filtration maintains dielectric cleanliness, which is crucial for cutting stability, surface integrity, and minimizing energy waste.

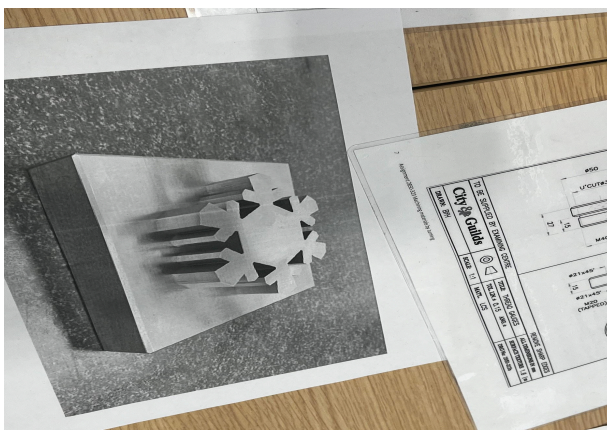


* wire EDM

cutting action.



* snowflake block.(pic from google)



Comparative tables and analysis.
Accuracy and surface finish comparison.

Process	Typical Tolerance	Surface Quality	Notes
Turning	± 0.05 to ± 0.10 mm	Moderate	Good for round profiles
Milling	± 0.05 mm	Moderate	Feature versatility
Drilling	± 0.05 mm	Rough to moderate	Improved with reaming
Broaching	± 0.01 to ± 0.05 mm	Good	Excellent for profiles
Honing	$< \pm 0.01$ mm	Very fine	Best for internal finish
Wire EDM	± 0.005 to ± 0.02 mm	Fine	Best for complex contours

Batch size and efficiency.

Process	Best Batch Sizes	Setup Time	Repeatability
Turning	Small	Short	Moderate
Milling	Small to Medium	Moderate	High
Drilling	Small	Very short	Moderate
Broaching	Medium to Large	High tooling	Very high
Honing	Small to Medium	Moderate	High
Wire EDM	Small to Medium	Moderate	Very high

Sustainability comparison.

Process	Energy Use	Fluid Use	Waste/Recyclability
Turning	Moderate	Low	Aluminium swarf recyclable
Milling	Medium to High	Moderate	Steel swarf recyclable
Drilling	Low	Minimal	Aluminium chips recyclable
Broaching	Medium	Minimal	Low waste
Honing	Moderate	Moderate	Abrasive residues
Wire EDM	High	High	Dielectric fluid management

Process selection rationale.

Each is based on a combination of accuracy requirements, batch size and sustainability.

1. Turning was selected for the aluminium shaft because it easily produces cylindrical features, tapers, and knurling with acceptable tolerances in low batch sizes and with recyclable waste.
2. Milling was chosen for the steel block because it can produce planar, curved, and deep features simultaneously with good precision.
3. Pillar drilling was chosen for the holework due to its efficiency for holes and threaded features in one-off or low batches.
4. Broaching was used for gear teeth because it produces consistent profiles efficiently in medium to higher volume runs.

5. Honing was selected for engine cylinder bores because it delivers superior surface finish and geometry control that other methods cannot achieve.
6. Wire EDM was best for the complex snowflake shape because it can cut intricate profiles with very high precision that traditional cutting cannot match.

Discussion.

This section delves into the strength and limitations of each process:

- Traditional methods are flexible and cost-efficient for general features and modest tolerances.
- Specialist methods justify their higher energy or tooling costs by delivering precision and finish that alternatives cannot.
- Sustainability varies by process, with traditional machining generating larger recyclable waste but lower energy usage, whereas specialist processes require careful fluid management but produce minimal bulk swarf.

Conclusion.

Secondary machining processes are essential to producing functional engineered components. Selecting the appropriate process depends on geometry, tolerance, production volume, and environmental impact.

Traditional methods suffice for straightforward features in smaller batches, while specialist machining is required for high precision or complex shapes. A balanced understanding of these factors leads to optimal manufacturing outcomes.

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